frameworks, sustainable resource management systems, and public cultural experiences.

9.4 Urban Air Mobility (UAM) and Drone Taxis

Urban air mobility (UAM) represents a new paradigm in transportation where autonomous and semi-autonomous aerial vehicles operate within city environments to transport passengers and goods. Enabled by advances in electric vertical take-off and landing (eVTOL) systems, lightweight composite airframes, high-density battery technologies, and AI-based flight control, UAM seeks to reduce traffic congestion, shorten travel times, and introduce sustainable alternatives to traditional road transport. Unlike conventional aircraft, UAM vehicles require minimal take-off space and can operate from compact hubs known as vertiports, making them suitable for dense metropolitan regions. Governments, industries, and smart city planners worldwide are preparing for UAM integration as part of next-generation multimodal mobility networks, where air and ground transport systems operate in synergy.

UAM development is shaped by three essential dimensions. First, the rise of passenger drones highlights the feasibility of short-range intra-city flights for airport transfers, emergency medical transport, and business district mobility. Second, cities must address infrastructure challenges, including vertiport placement, safe aerial corridor design, air traffic management, noise zoning, and charging logistics. Third, successful deployment requires robust regulatory frameworks governing airworthiness certification, autonomous operation permissions, flight safety protocols, and cybersecurity standards, alongside securing public trust and community acceptance.

Together, these developments signal a transformative shift in how cities may function in the near future where the sky evolves from passive space